DABs in DATABRICKS

Presentation

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INTRODUCTION

Databricks Asset Bundles (DABs)

- A way to define Databricks resources using code
- Uses YAML configuration files
- Enables automated deployment
- Supports version control with Git
- Designed for production-ready workflows

Databricks
Asset
Bundles

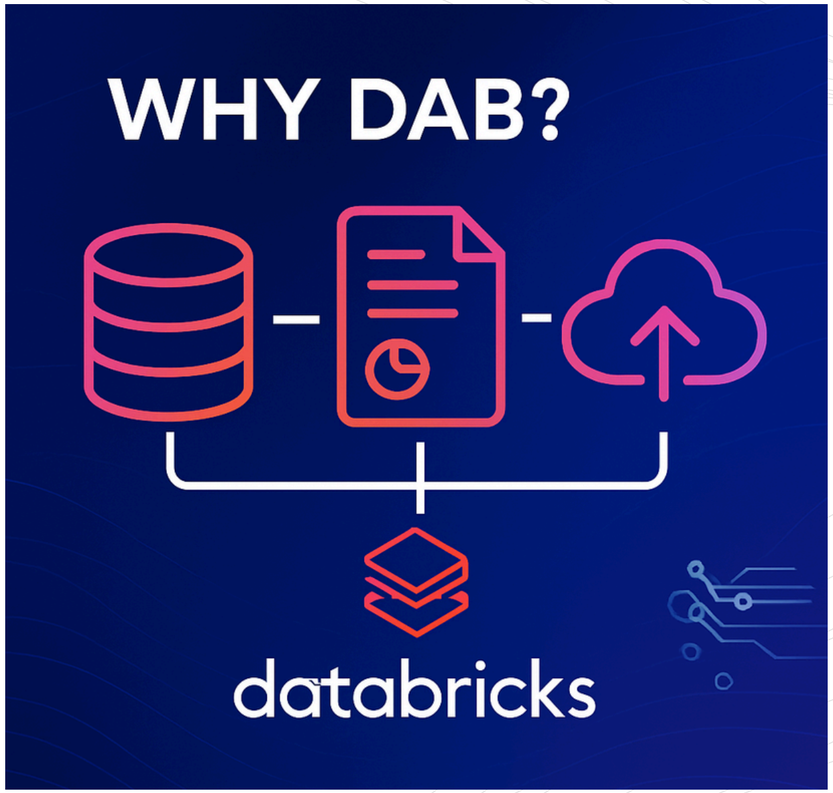
WHY DAB?

Problems Without DABs:

- Manual job creation in UI
- Hard to track changes
- Difficult to manage Dev, Test, Prod
- Risky production updates

Solution:

DABs provide structured, repeatable, and automated deployment across environments.

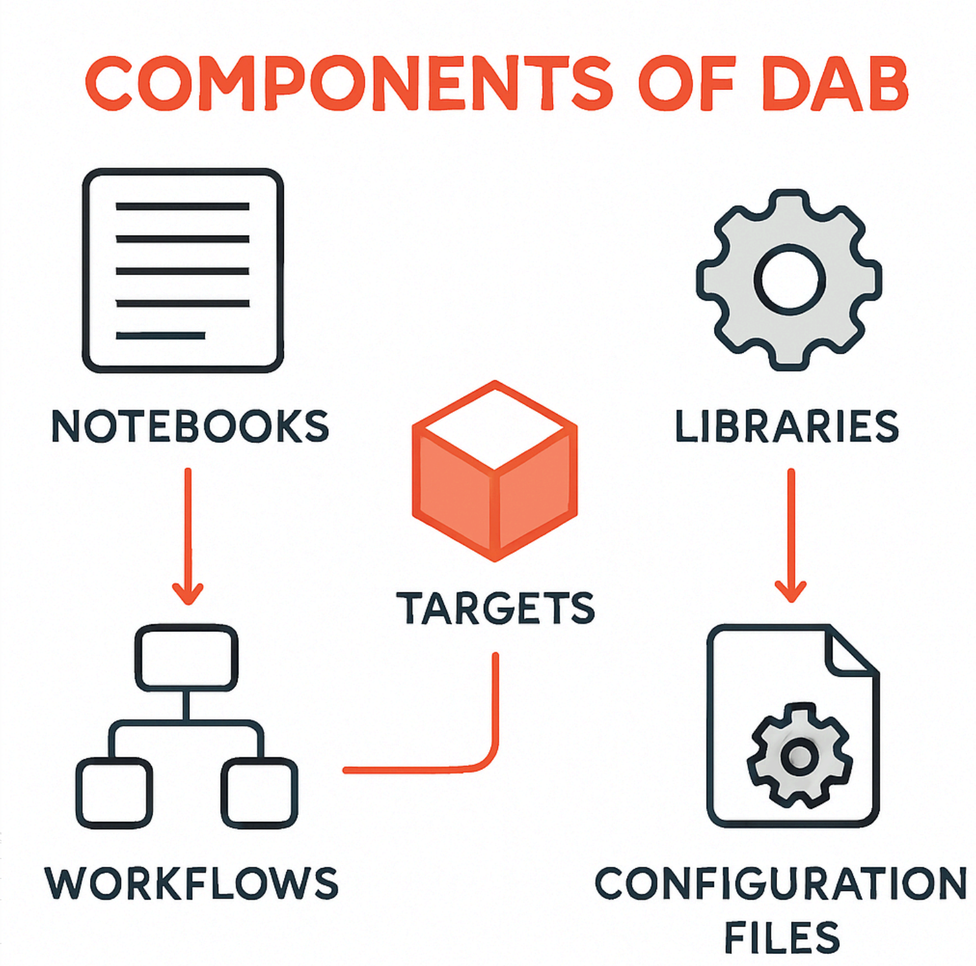


COMPONENTS OF DAB

A typical DAB project contains:

- databricks.yml – Main configuration file
- Resources folder – Jobs, workflows, pipelines
- Source code – Notebooks / Python files
- Targets – Dev, Test, Prod environments

Everything that can be created in UI can be defined in YAML.



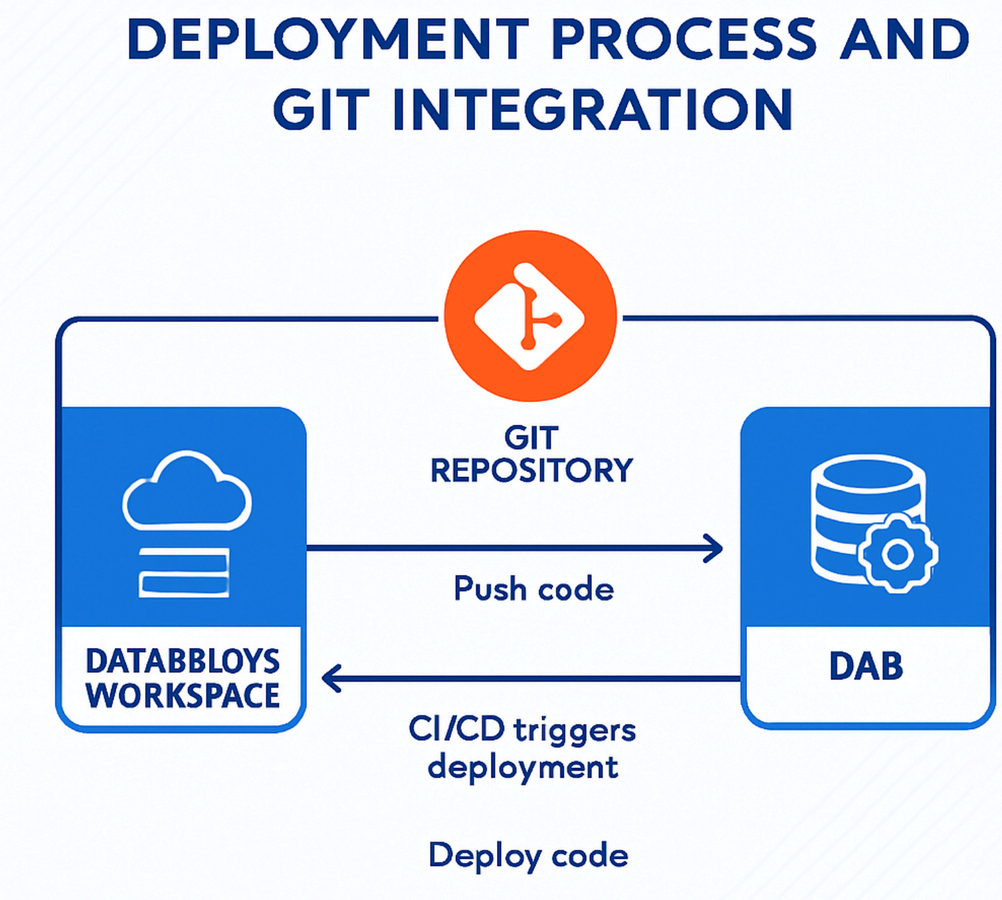
DEPLOYMENT PROCESS AND GIT INTEGRATION

Using Databricks CLI:

- bundle init → Create project
- bundle validate → Validate config
- bundle deploy → Deploy resources
- bundle run → Execute job

With Git Integration:

- Write code locally
- Push to GitHub
- CI/CD triggers deployment
- Workspace updates automatically



REAL-WORLD USE CASE

Key Benefits:

- Infrastructure as Code
- Version controlled
- Easy Dev → Test → Prod migration
- CI/CD integration
- Reproducible deployments
- Team collaboration friendly

Example Use Case:

Daily ETL Pipeline

- Read raw data
- Transform
- Store in Delta tables
- Run on schedule
- Defined once in YAML → deployed anytime.

THANK YOU!